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Benchmarking multiblock methods with canonical factorization

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ABSTRACT

Data measured on the same observations and organized in blocks of variables — from different measurement sources or deduced from topics specified by the user — are common in practice. Multiblock exploratory methods are useful tools to extract information from data in a reduced and interpretable common space. However, many methods have been proposed independently and the users are often lost in selecting the appropriate one, especially as they do not always lead to the same results or because outputs do not have the same form. For this purpose, the data decomposition by canonical factorization was introduced thus applied to some widely-used methods, CPCA, MCOA, MFA, STATIS and CCSWA. The methods were compared on simulated (resp. real) data whose structure is controlled (resp. known). Theoretical and practical results pinpoint that the block-structure must be carefully explored beforehand. The number of block-variables and the block-variance distribution along dimensions impacts the choice of the block-scaling. The observation-structure within and between blocks impacts the choice of the method. CPCA or MCOA mix common and specific information, STATIS highlights common structure only whereas CCSWA focuses on specific information. To enable these diagnoses, methods and proposed comparison tools are available on R, `Matlab` or `Galaxy`.

1. Introduction

Data sets organized in blocks of variables measured on the same observations are common in practice. Blocks may correspond to (quantitative) variables from different measurement sources or to topics specified by the user. For example in omics data, blocks consist of variables obtained from different techniques (e.g., proteomics, metabolomics). In food science, relationships between physico-chemical measurements, microbiological characterization and sensory attributes can be explored. The aims of exploratory analysis of multiblock data are manifold: (i) jointly reduce the dimensions of multiple blocks, (ii) investigate relationships between blocks, (iii) and between variables within and between blocks, and (iv) recover within-block variation to highlight common and specific block-structure in a common space. For this purpose, multiblock exploratory methods are appropriate tools. Many methods have been proposed independently in the literature within different application frameworks: Generalized Canonical Correlation Analysis (GCCA) [1] and its popular case SUMCOR [2], Generalized Procruste Analysis (GPA) [3] MAXBET [4] Consensus Principal Component Analysis (CPCA) [5], Multiblock Principal Component Analysis (MBPCA) [6], Multiple Factorial Analysis (MFA) [7], Structuration de Tableaux A Trois Indices de la Statistique (STATIS) [8], Multiple CO-inertia Analysis (MCOA) [9], Hierarchical Principal Component Analysis (HPCA) [10,11], COMDIM also known as Common Components and Specific Weights Analysis (CCSWA) [12], or SUM-PCA [13]. Users are often lost in choosing the appropriate method because outputs do not have the same form or do not lead to the same results.

Compared to the number of methods, their interconnection has been little investigated. Some of them (e.g., MBPCA, HPCA) were not described with a criterion but with an iterative algorithm, while some others (e.g., MCOA, MFA) derived from eigendecomposition. An integrative analysis of some methods (SUM-PCA, PCovR, MFA, STATIS, SCA-P) based on Simultaneous Component Analysis (SCA) has been proposed [14]. A monotonicity property of HPCA was revealed and an optimization criterion was presented to show equivalence between HPCA and CCSWA [15]. A new formulation of CCSWA was introduced with a criterion similar to that of MCOA or CPCA [16]. New properties of CPCA revealed its connection with MCOA and PCA [17]. Some methods (e.g., GCCA, SUMCOR) can be considered as special cases

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of Regularized Generalized Canonical Correlation Analysis (RGCCA) when the concatenated block is considered as the dependent one [18]. GCCA, CCSWA and HPCA were considered in a unified framework [19]. Despite these clarifications, users still need to compare methods theoretically and, most importantly, get a practical guide to choose the appropriate method.

Our first aim is to reformulate the outputs of widely-used methods, CPCA, MCOA, MFA, STATIS and CCSWA (equivalent to COMDIM, ACCPS, HPCA). For this purpose, we proposed to introduce data decomposition by canonical factorization to each method. Benefits are two-fold: (i) standardization of method outputs — comparable to those of standard PCA — and (ii), for a given method, relation between overall- and block outputs. Our second aim is to simulate different data and compare multiblock methods on these outputs. Without any relevant model, it is not possible to simulate multiblock data properly. Thanks to canonical factorization, it is. Our final aim is to apply multiblock methods to real data in order to move towards a clear user guideline. The rest of this article is organized as follows. In Section 2, the notion of canonical factorization is given (Section 2.2) and then applied to multiblock methods (Section 2.3). The way to simulate and compare methods is given (Section 2.4). In Section 3, multiblock methods were compared on a simulation study (Section 3.1) and on data pertaining to multiple data integration for food (Section 3.2). In Section 4, results are summarized and perspectives for future work are drawn.

2. Method

2.1. Notations

Matrices are denoted by bold upper-case letters ($\mathbf{X}$) and vectors by bold lower-case letters ($\mathbf{x}$). $\mathbf{X}^T$ denotes the transpose operation of a matrix $\mathbf{X}$. For a square matrix $\mathbf{X}$, $\mathrm{trace}(\mathbf{X})$ is the sum of diagonal elements of $\mathbf{X}$ and $\mathrm{diag}(\mathbf{x})$ is a diagonal matrix whose diagonal elements are elements of $\mathbf{x}$. $\|\mathbf{X}\|$ is the Frobenius norm of $\mathbf{X}$. Multiblock data are described with N observations and K blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ of $(J_1, \ldots, J_K)$ variables, with $J = \sum_{k=1}^{K} J_k$. The concatenated data is $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K]$, and matrix rank is $\mathrm{rank}(\mathbf{X}) = H$. Without loss of generality, variables are assumed to be column-centred. The concatenated data can be decomposed with components and loading by $\mathbf{X} = \tilde{\mathbf{T}}\tilde{\mathbf{P}}^T$ (PCA) or with standardized ones by $\mathbf{X} = \mathbf{TDP}^T$ (SVD). Let $\mathbf{T} = [\mathbf{t}^{(1)}, \ldots, \mathbf{t}^{(H)}]$ be standardized common components of size $(N \times H)$ ($\|\mathbf{t}^{(h)}\| = 1$ for $h = 1, \ldots, H$ with $\mathbf{t}^{(h)}\mathbf{t}^{(h')T} = 0$ for $h = 1, \ldots, H$, $h' = 1, \ldots, H$, $h \neq h'$), $\mathbf{D}$ the diagonal matrix of scaling of size $(H \times H)$ and $\mathbf{P} = [\mathbf{p}^{(1)}, \ldots, \mathbf{p}^{(H)}]$ standardized common loadings of size $(J \times H)$ ($\|\mathbf{p}^{(h)}\| = 1$ for $h = 1, \ldots, H$).

2.2. Canonical factorization of multiblock data

Rationale. The canonical factorization of data by a multiblock method consists of assigning a standardized decomposition to each block, concatenated data included. Canonical factorization is the decomposition of $(K + 1)$ blocks by a product of matrices with various constraints, depending on multiblock methods. This seek a common mathematical concept, of which each multiblock method — associated with a criterion to be optimized — is a particular implementation associated with unique data decompositions. The originality of canonical factorization is manifold: (i) it decomposes multiblock data into common and thus comparable parameters, (ii) it highlights relationships between overall analysis (decomposition of $\mathbf{X}$) and block-analyses (decompositions of $\mathbf{X}_1, \ldots, \mathbf{X}_K$), (iii) common and block-parameters have a statistical and geometrical interpretation which clarifies the strategy adopted by methods for analysing multiblock data.

Proposal. The canonical factorization of multiblock data is the decomposition of the $(K + 1)$ blocks — $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ and $\mathbf{X}$ — following:

$$\begin{cases} \mathbf{X} = \mathbf{TDP}^T \\ \mathbf{X}_k = \mathbf{TD}_k\mathbf{P}_k^T \quad \text{for} \quad k = (1, \ldots, K) \\ \text{with the constraints} \quad \mathbf{T}^T\mathbf{T} = \mathbf{I} \quad \text{and} \quad \|\mathbf{p}^{(h)}\| = \|\mathbf{p}_k^{(h)}\| = 1 \\ \quad \text{for} \quad h = (1, \ldots, H)\,, \end{cases} \tag{1}$$

with $\mathbf{T}$ the standardized and orthogonal common components, $\mathbf{P}$ the standardized (not necessarily orthogonal) common loadings, $\mathbf{P}_k$ the standardized block-loadings, $\mathbf{D}$ and $\mathbf{D}_k$ the diagonal scaling matrices with property $\mathbf{D}^2 = \sum_{k=1}^{K} \mathbf{D}_k^2$ that relate common and block-analyses.

Proof. The overall data are decomposed into $\mathbf{X} = \tilde{\mathbf{T}}\tilde{\mathbf{P}}^T$, e.g., with a PCA. The two matrices are standardized following $\tilde{\mathbf{T}} = \mathbf{TD}_T$ and $\tilde{\mathbf{P}}^T = \mathbf{D}_P\mathbf{P}^T$, the scaling matrices $\mathbf{D}_T$ and $\mathbf{D}_P$ being both diagonal of size $(H \times H)$. The canonical factorization of $\mathbf{X}$ is given by:

$$\begin{aligned} &\mathbf{X} = \mathbf{T}\left(\mathbf{D}_T\mathbf{D}_P\right)\mathbf{P}^T = \mathbf{TDP}^T \\ &\text{with the constraints} \quad \mathbf{T}^T\mathbf{T} = \mathbf{I} \quad \text{and} \quad \|\mathbf{p}^{(h)}\| = 1 \\ &\quad \text{for} \quad h = (1, \ldots, H)\,, \end{aligned} \tag{2}$$

with $\mathbf{D} = \mathbf{D}_T\mathbf{D}_P$. If $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K]$, it follows $\mathbf{X} = \tilde{\mathbf{T}}[\tilde{\mathbf{P}}_1^T | \ldots | \tilde{\mathbf{P}}_K^T]$ with $\tilde{\mathbf{P}}^T = [\tilde{\mathbf{P}}_1^T | \ldots | \tilde{\mathbf{P}}_K^T]$, the non-standardized loading matrix inheriting data structure of $\mathbf{X}$. It derives $\mathbf{X}_k = \tilde{\mathbf{T}}\tilde{\mathbf{P}}_k^T$, blocks being decomposed into the same orthogonal basis $\tilde{\mathbf{T}}$. Standardizations are also applied to $\mathbf{X}_k = \mathbf{T}\left(\mathbf{D}_T\mathbf{D}_{P_k}\right)\mathbf{P}_k^T$, the $(K+1)$ scaling matrices $\mathbf{D}_T$ and $(\mathbf{D}_{P_1}, \ldots, \mathbf{D}_{P_K})$ being all diagonal of size $(H \times H)$. The canonical factorization of each block $\mathbf{X}_k$ is given by:

$$\begin{aligned} &\mathbf{X}_k = \mathbf{T}\left(\mathbf{D}_T\mathbf{D}_{P_k}\right)\mathbf{P}_k^T = \mathbf{TD}_k\mathbf{P}_k^T \\ &\text{with the constraints} \quad \mathbf{T}^T\mathbf{T} = \mathbf{I} \quad \text{and} \quad \|\mathbf{p}_k^{(h)}\| = 1 \\ &\quad \text{for} \quad h = (1, \ldots, H),\ k = (1, \ldots, K)\,, \end{aligned} \tag{3}$$

with $\mathbf{D}_k = \mathbf{D}_T\mathbf{D}_{P_k}$. To demonstrate relationships between $\mathbf{D}$ and $\mathbf{D}_k$:

$$\begin{aligned} &\mathbf{X} = \mathbf{TD}_T[\tilde{\mathbf{P}}_1^T | \ldots | \tilde{\mathbf{P}}_K^T] \\ &\mathbf{X} = \mathbf{TD}\left[\left(\mathbf{D}_T\mathbf{D}^{-1}\mathbf{D}_1\right)\mathbf{P}_1^T | \ldots | \left(\mathbf{D}_T\mathbf{D}^{-1}\mathbf{D}_K\right)\mathbf{P}_K^T\right] \end{aligned}$$

Because $\|\mathbf{p}^{(h)}\| = 1$, it follows for a given dimension h, $\|\mathbf{p}\|^2 = \mathbf{p}^T\mathbf{p} = \sum_{k=1}^{K}\left(\frac{\mathbf{d}_T\mathbf{d}_k}{\mathbf{d}}\right)^2 \mathbf{p}_k^T\mathbf{p}_k$. Because $\|\mathbf{p}_k^{(h)}\| = 1$, it follows $\sum_{k=1}^{K}\left(\mathbf{D}_T\mathbf{D}^{-1}\mathbf{D}_k\right)^2 = \mathbf{I}$, then $\mathbf{D}^2 = \sum_{k=1}^{K}\left(\mathbf{D}_T\mathbf{D}_k\right)^2$. For normalized common components ($\|\mathbf{t}^{(h)}\| = 1$ for $h = 1, \ldots, H$), $\mathbf{D}^2 = \sum_{k=1}^{K}\mathbf{D}_k^2$.

Interpretation. For a given method, canonical factorization allows us to project the $\mathbf{X}_k$ columns into a common space spanned by the (orthogonal and normalized) common components $\mathbf{T}$, considered as a common model for blocks. Common and block parameters are related with each others. Because canonical factorization of data by multiblock methods have the same normalized format (Eq. (1)), methods can be compared on the same parameters (i.e., $\mathbf{T}$, $\mathbf{D}_k$). The property $\mathbf{D}^2 = \sum_k \mathbf{D}_k^2$ means that the variance of concatenated data is the sum of block-variances.

2.3. Canonical factorization of data by multiblock methods

The canonical factorization of concatenated multiblock data by PCA is introduced as a reference. Then, multiblock methods are presented by: (i) their original algorithm or criterion, (ii) a criterion related to PCA of concatenated data, and (iii) their canonical factorization.

2.3.1. PCA of concatenated data

The PCA of $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K]$ — with standardized components — is based, for the first dimension, on criterion [20]:

$$\underset{\mathbf{t}, \mathbf{w}_1, \ldots, \mathbf{w}_K}{\arg\max} \sum_{k=1}^{K} \mathrm{cov}^2(\mathbf{t}, \mathbf{X}_k\mathbf{w}_k) \quad \text{with the constraints} \quad \|\mathbf{t}\| = \|\mathbf{w}_k\| = 1\,. \tag{4}$$

The solution is given by $\mathbf{w}_k = \frac{\mathbf{X}_k^T\mathbf{t}}{\|\mathbf{X}_k^T\mathbf{t}\|}$. While replacing $\mathbf{w}_k$ in Eq. (4), $\mathbf{T}$ are either (normalized) eigenvectors of $(\mathbf{X}\mathbf{X}^T)$ or can be sought by the deflation of $\mathbf{X}_k$ into successive components $\mathbf{t}$. Equivalently, $\mathbf{t}$ is the first left singular vectors of $\mathbf{X}$ and $\mathbf{w} = [\mathbf{w}_1|\dots|\mathbf{w}_K]$ — with $\mathbf{t} = \mathbf{X}\mathbf{w}$ — is deduced from the right singular vectors of $\mathbf{X}$ up to a block-scaling. Therefore, the decomposition of $\mathbf{X}_k$ onto the orthogonal basis is given by $\mathbf{X}_k = \mathbf{X}_k^{(1)}$ and $\mathbf{X}_k^{(h+1)} = \left[\mathbf{I} - \mathbf{t}^{(h)}\mathbf{t}^{(h)T}\right]\mathbf{X}_k^{(h)}$ for each dimension $h = (1,\dots,H)$. It derives:

$$\begin{array}{rcll} \mathbf{X}_k & = & \sum_{h=1}^{H}\mathbf{t}^{(h)}\mathbf{t}^{(h)T}\mathbf{X}_k & \text{for} \quad k=(1,\dots,K) \\ \Leftrightarrow \quad \mathbf{X}_k & = & \mathbf{T}\mathbf{T}^T\mathbf{X}_k & \\ \Leftrightarrow \quad \mathbf{X}_k & = & \mathbf{T}\mathbf{D}_k\mathbf{W}_k^T & \end{array} \tag{5}$$

with $\mathbf{D}_k$ the scaling matrix of $\mathbf{W}_k$. From Eqs. (2) and (3), it follows:

$$\begin{cases} \mathbf{T} & = [\mathbf{t}^{(1)},\dots,\mathbf{t}^{(H)}] & \\ \mathbf{D}_k & = \mathrm{diag}\left(\|\mathbf{X}_k^T\mathbf{t}^{(1)}\|,\dots,\|\mathbf{X}_k^T\mathbf{t}^{(H)}\|\right) & \text{for} \quad k=(1,\dots,K) \\ \mathbf{P}_k^T & = [\mathbf{w}_k^{(1)T},\dots,\mathbf{w}_k^{(H)T}] & \text{for} \quad k=(1,\dots,K) \\ \mathbf{D} & = \mathrm{diag}\left(\sqrt{\sum_k\|\mathbf{X}_k^T\mathbf{t}^{(1)}\|^2},\dots,\sqrt{\sum_k\|\mathbf{X}_k^T\mathbf{t}^{(H)}\|^2}\right) & \\ \mathbf{P}^T & = \left[\mathbf{D}_1\mathbf{D}^{-1}\mathbf{P}_1^T|\dots|\mathbf{D}_K\mathbf{D}^{-1}\mathbf{P}_K^T\right] . & \end{cases} \tag{6}$$

All multiblock methods look for common components $\mathbf{T}$, orthogonal to each other, on which blocks $(\mathbf{X}_1,\dots,\mathbf{X}_K)$ are decomposed. Diagonal elements of $\mathbf{D}_k$, $\|\mathbf{X}_k^T\mathbf{t}\|$ — scaling parameters of block-loadings $\mathbf{W}_k$ — are interpreted as block-variance explained by common components. It helps us to understand how multiblock methods partition block-variability along dimensions of common space. The diagonal elements of $\mathbf{D}$ are interpreted as the total variance explained by common component.

2.3.2. Consensus Principal Component Analysis (CPCA)

CPCA has been proposed by [5] in chemometrics by extending NIPALS algorithm to several blocks. The algorithm proceeds in two steps. First, the parameters — common components, common loadings, block-components and block-loadings — are computed according to a procedure repeated until convergence. A standardized (random) vector $\mathbf{t}$ is considered as a starting point. The columns in $\mathbf{X}_k$ are regressed on $\mathbf{t}$, leading to block-loadings $\mathbf{w}_k^T = (\mathbf{t}^T\mathbf{t})^{-1}\mathbf{t}^T\mathbf{X}_k$ in turn standardized. The block-components are computed by $\mathbf{t}_k = \mathbf{X}_k\mathbf{w}_k$. The common axis is computed by $\boldsymbol{\omega}^T = (\mathbf{t}^T\mathbf{t})^{-1}\mathbf{t}^T[\mathbf{t}_1|\dots|\mathbf{t}_K]$, thus composed of K vectors $\omega_k^T = (\mathbf{t}^T\mathbf{t})^{-1}\mathbf{t}^T\mathbf{t}_k$. The common component is updated by setting $\mathbf{t} = [\mathbf{t}_1|\dots|\mathbf{t}_K]\boldsymbol{\omega}^T$. After convergence and in a second step, higher-order parameters are computed by deflating $\mathbf{X}_k$ blocks with respect to common components. The procedure has monotonicity properties [17] and optimizes the PCA problem (4). This leads to canonical factorization of $\mathbf{X}$ and $(\mathbf{X}_1,\dots,\mathbf{X}_K)$ by CPCA according to Eq. (6).

2.3.3. Multiple CO-inertia Analysis (MCOA)

In ecological field, MCOA has been proposed [9,17] as an alternative to MAXVAR [1]. It is based on the same problem as CPCA (Eq. (4)). For the first-order solution, the parameters $\mathbf{w}_k$ and $\mathbf{t}$ are the same as the ones of CPCA. The difference between MCOA and CPCA is the deflation step, with respect to block-loadings $\mathbf{w}_k$ (MCOA) and to common component $\mathbf{t}$ (CPCA). The common components are orthogonal to each other (MCOA and CPCA). Due to block-deflation, the maximum MCOA dimension is $H' = \min\left[\mathrm{rank}(\mathbf{X}_1,\dots,\mathbf{X}_K)\right]$. Without any loss of generality but keeping in mind that $\mathbf{X}$ and $\mathbf{X}_k$ reconstructions are not complete and contain residual terms, canonical factorization of $(K+1)$ data sets is performed on a H'-dimensional space:

$$\begin{array}{rcll} \mathbf{X}_k & = & \sum_{h=1}^{H'}\mathbf{t}^{(h)}\mathbf{t}^{(h)T}\mathbf{X}_k + \mathbf{R}_k & \text{for} \quad k=(1,\dots,K) \\ \mathbf{X} & = & \sum_{h=1}^{H'}\mathbf{t}^{(h)}\mathbf{t}^{(h)T}\mathbf{X} + \mathbf{R} & \end{array}$$

Therefore, canonical factorization of $\mathbf{X}$ and $(\mathbf{X}_1,\dots,\mathbf{X}_K)$ by MCOA is given by the same parameters as CPCA (Eq. (6)) but in a H'-dimension space.

2.3.4. Multiple Factorial Analysis (MFA)

MFA has been proposed by [7] and is wildly used in sensometrics [21]. It is known as a PCA of concatenated blocks, each of which being scaled with the inverse of $\sqrt{\lambda_k^{(1)}}$, the square root of the first eigenvalue of block-PCA, namely $\mathbf{X}_\lambda = [(1/\sqrt{\lambda_k^{(1)}})\mathbf{X}_1|\dots|(1/\sqrt{\lambda_K^{(1)}})\mathbf{X}_K]$. This block-scaling gives more weight to blocks with the lowest within-block correlation. MFA (with standardized components) can be presented with following criterion:

$$\underset{\mathbf{t},\mathbf{w}_1,\dots,\mathbf{w}_K}{\arg\max}\sum_{k=1}^{K}\mathrm{cov}^2\left(\mathbf{t},\frac{1}{\sqrt{\lambda_k^{(1)}}}\mathbf{X}_k\mathbf{w}_k\right) \quad \text{with the constraints} \quad \|\mathbf{t}\| = \|\mathbf{w}_k\| = 1 . \tag{7}$$

The solution is given by $\mathbf{w}_k = \frac{1/\sqrt{\lambda_k^{(1)}}\mathbf{X}_k^T\mathbf{t}}{\|1/\sqrt{\lambda_k^{(1)}}\mathbf{X}_k^T\mathbf{t}\|} = \frac{\mathbf{X}_k^T\mathbf{t}}{\|\mathbf{X}_k^T\mathbf{t}\|}$. Because MFA is a PCA of weighted concatenated matrix $\mathbf{X}_\lambda$, $\mathbf{T}$ are eigenvectors of $\left(\sum_{k=1}^{K}\frac{1}{\lambda_k^{(1)}}\mathbf{X}_k\mathbf{X}_k^T\right)$. From Eq. (7) — H being the rank of $\left(\sum_{k=1}^{K}\frac{1}{\lambda_k^{(1)}}\mathbf{X}_k\mathbf{X}_k^T\right)$ — canonical factorization of $\mathbf{X}$ and $(\mathbf{X}_1,\dots,\mathbf{X}_K)$ by MFA is given by Eq. (6).

2.3.5. Structuration de Tableaux A Trois Indices de la Statistique (STATIS)

STATIS has been proposed by [8] and is also widely used in sensometrics where a compromise (between judges) is mainly sought. The method assumes that information behind K blocks is contained in K matrices of scalar products $(\mathbf{X}_1\mathbf{X}_1^T,\dots,\mathbf{X}_K\mathbf{X}_K^T)$. The method is based on a two-step procedure. The first (inter-structure) one searches for a compromise matrix denoted $\mathbf{S}$, solution of:

$$\underset{\mathbf{S},\alpha_1,\dots,\alpha_K}{\arg\min}\sum_{k=1}^{K}\|\mathbf{X}_k\mathbf{X}_k^T - \alpha_k\mathbf{S}\|^2 \quad \text{with the constraint} \quad \sum_{k=1}^{K}\alpha_k^2 = 1 . \tag{8}$$

The solution is given by $\mathbf{S} = \sum_{k=1}^{K}\alpha_k\mathbf{X}_k\mathbf{X}_k^T$ with $\boldsymbol{\alpha} = (\alpha_1,\dots,\alpha_K)$ the first eigenvector of $\mathbf{C} = [\mathrm{trace}(\mathbf{X}_k\mathbf{X}_k^T\mathbf{X}_{k'}\mathbf{X}_{k'}^T)/\sqrt{\mathrm{trace}(\mathbf{X}_k\mathbf{X}_k^T)\,\mathrm{trace}(\mathbf{X}_{k'}\mathbf{X}_{k'}^T)}]$ of dimension $(K\times K)$ [22–24]. The second (compromise) step consists in looking for common components $\mathbf{T}$ from the eigen decomposition of $\mathbf{S}$. Because $\mathbf{S} = \sum_{k=1}^{K}\alpha_k\mathbf{X}_k\mathbf{X}_k^T$ — with $\alpha_k \geq 0$ [25] — it follows $\mathbf{S} = \mathbf{X}_\alpha\mathbf{X}_\alpha^T$ with concatenated block-weighted matrix $\mathbf{X}_\alpha = [\sqrt{\alpha_1}\mathbf{X}_1|\dots|\sqrt{\alpha_K}\mathbf{X}_K]$. Then, common (normalized) components $\mathbf{T}$ can be found by PCA of $\mathbf{X}_\alpha$:

$$\underset{\mathbf{t},\mathbf{w}_1,\dots,\mathbf{w}_K}{\arg\max}\sum_{k=1}^{K}\mathrm{cov}^2\left(\mathbf{t},\sqrt{\alpha_k}\mathbf{X}_k\mathbf{w}_k\right) \quad \text{with the constraints} \quad \|\mathbf{t}\| = \|\mathbf{w}_k\| = 1 . \tag{9}$$

The solution is given by $\mathbf{w}_k = \frac{\sqrt{\alpha_k}\mathbf{X}_k^T\mathbf{t}}{\|\sqrt{\alpha_k}\mathbf{X}_k^T\mathbf{t}\|} = \frac{\mathbf{X}_k^T\mathbf{t}}{\|\mathbf{X}_k^T\mathbf{t}\|}$ and $\mathbf{T}$ the eigenvectors of $\left(\sum_{k=1}^{K}\alpha_k\mathbf{X}_k\mathbf{X}_k^T\right)$. From Eq. (9) — H being the rank of $\left(\sum_{k=1}^{K}\alpha_k\mathbf{X}_k\mathbf{X}_k^T\right)$ assuming non-trivial cases where $\alpha_k \neq 0$ — canonical factorization of $\mathbf{X}$ and $(\mathbf{X}_1,\dots,\mathbf{X}_K)$ by STATIS is given by Eq. (6).

Block-weight proportionality property. From Eq. (8) and eigen decomposition of $\mathbf{S}$, following equality holds:

$$(H-1)\sum_{k=1}^{K}\|\mathbf{X}_k\mathbf{X}_k^T\|^2 + \sum_{k=1}^{K}\|\mathbf{X}_k\mathbf{X}_k^T - \alpha_k\mathbf{S}\|^2 = \sum_{k=1}^{K}\sum_{h=1}^{H}\|\mathbf{X}_k\mathbf{X}_k^T - \alpha_k\lambda^{(h)}\mathbf{t}^{(h)}\mathbf{t}^{(h)^T}\|^2 .$$

For a given dimension h and a given block $\mathbf{X}_k$, its block-weight is equal to $\alpha_k\lambda^{(h)}$, with $\lambda^{(h)}$ the hth eigenvalue associated with eigen decomposition of $\left(\sum_{k=1}^{K}\alpha_k\mathbf{X}_k\mathbf{X}_k^T\right)$. This means that for a given block $\mathbf{X}_k$, its H block-weights $(\alpha_k\lambda^{(1)},\dots,\alpha_k\lambda^{(H)})$ are proportional along dimensions.

This property is important to be known by the user because it has strong practical consequences further illustrated in the simulation study.

2.3.6. Common components and specific weights analysis (CCSWA)

CCSWA has been proposed as an alternative to STATIS while relaxing block-weight proportionality property along dimensions, and used in sensometrics for analysis of standard or free profile data [12,26–28]. For a given dimension, CCSWA determines a common component $\mathbf{t}$ and K block-saliences α_k, solutions of problem:

$$\underset{\mathbf{t},\alpha_1,\ldots,\alpha_K}{\arg\min} \sum_{k=1}^{K} \|\mathbf{X}_k\mathbf{X}_k^T - \alpha_k\mathbf{t}\mathbf{t}^T\|^2 \quad \text{with the constraint} \quad \|\mathbf{t}\| = 1\ . \tag{10}$$

The solution is solved by an Alternating Least Square algorithm reiterated until convergence of criterion (10). In the first stage, a standardized (random) salience vector $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)$ is considered. The common component $\mathbf{t}$ is computed as the first normalized eigenvector of $\sum_{k=1}^{K} \alpha_k\mathbf{X}_k\mathbf{X}_k^T$. The block-saliences are updated by $\alpha_k = \|\mathbf{X}_k^T\mathbf{t}\|^2$ for $k = (1, \ldots, K)$. After convergence and in a second stage, higher-order parameters are computed by deflating $\mathbf{X}_k$ blocks with respect to common components.

Property. CCSWA's saliences are different from the weights α_k computed in STATIS — and not necessarily proportional — from one dimension to another because CCSWA is based, for a given dimension h, on problem:

$$\underset{\mathbf{t},\alpha_1,\ldots,\alpha_K}{\arg\min} \sum_{k=1}^{K} \|\mathbf{X}_k^{(h)}\mathbf{X}_k^{(h)T} - \alpha_k^{(h)}\mathbf{t}^{(h)}\mathbf{t}^{(h)T}\|^2 \quad \text{with the constraint} \quad \|\mathbf{t}^{(h)}\| = 1\ , \tag{11}$$

the saliences being derived from common components by $\alpha_k^{(h)} = \|\mathbf{X}_k^{(h)T}\mathbf{t}^{(h)}\|^2$.

Hanafi and Qannari [16] demonstrated that CCSWA optimizes problem:

$$\underset{\mathbf{t},\mathbf{w}_1,\ldots,\mathbf{w}_K}{\arg\max} \sum_{k=1}^{K} \mathrm{cov}^4\left(\mathbf{t}, \mathbf{X}_k\mathbf{w}_k\right) \quad \text{with the constraints} \quad \|\mathbf{t}\| = \|\mathbf{w}_k\| = 1\ . \tag{12}$$

The solution is given by $\mathbf{w}_k = \frac{\mathbf{X}_k^T\mathbf{t}}{\|\mathbf{X}_k^T\mathbf{t}\|}$ and $\mathbf{T}$ the eigenvectors of $\sum_{k=1}^{K} \alpha_k\mathbf{X}_k\mathbf{X}_k^T$. The saliences are *a posteriori* derived from $\alpha_k = \|\mathbf{X}_k^T\mathbf{t}\|^2$. From Eq. (12), canonical factorization of $\mathbf{X}$ and $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ by CCSWA is presented by Eq. (6). Details and proofs are given in [16], especially Appendix 4.

2.3.7. Summary

All of the methods examined can be decomposed in the same way using Eq. (1). However, they summarize information provided by blocks in different ways. Their parameters $\mathbf{T}$, $\mathbf{P}_k$, $\mathbf{D}_k$, $\mathbf{P}$ and $\mathbf{D}$, differ from one method to another, due to space deformation and specific rotations of $\mathbf{T}$ in $\mathbb{R}^N$ space. From Eqs. (4), (7), (9), (12), these differences result from: (i) specific block-weightings (for MFA and STATIS), (ii) covariance power in criterion (power 4 for CCSWA and 2 otherwise) and (iii) deflation procedure (on $\mathbf{w}_k$ for MCOA and on $\mathbf{t}$ otherwise). All these exploratory multiblock methods can be viewed as special cases of rGCCA with specific options and while considering concatenated dataset $\mathbf{X}$ to be explained (i.e.,'super-block') [18,29]. Some details are given in Appendix A.1. The theoretical specificities have practical consequences that lead users to clarify two issues before choosing multiblock method. These issues derive from block structure which must be explored beforehand, e.g., with K block-PCA. (i) The first issue is related to variable-structure and concerns block-variance distribution along dimensions. If block(s) with a scattered variance along dimensions are more important than others, then MFA — or another method with a MFA-like block-scaling — should be chosen. Otherwise, a standard block-scaling (e.g., with inertia) should be applied. (ii) The second issue is related to observation-structure. If the aim is to emphasize only common structure, STATIS — and its block-scaling identical along dimensions (Eq. (9)) — should be chosen. If the aim is to consider what is specific to each block along dimensions, CCSWA — and its specific block-scaling (Eq. (11)) — should be chosen. Otherwise, if the aim is to mix common and specific information along dimensions, CPCA or MCOA (Eq. (4)) should be chosen, the latter method giving a more condensed information (Section 2.3.3).

2.4. Comparison of multiblock exploratory methods

How to compare multiblock methods? In case of multiblock data, the common structure to all blocks is observation-structure, variables being different from one block to another. The components $\mathbf{T}$ represent this common structure. In a linear approach, there is a duality between observation- and variable-structure. This can be shown with the following equivalent canonical factorizations of $\mathbf{X}_k$, while taking into account that components and loadings are normalized and respectively orthogonal to each other ($\mathbf{T}^T\mathbf{T} = \mathbf{P}_k^T\mathbf{P}_k = \mathbf{I}$):

$$\begin{aligned} \mathbf{X}_k &= \mathbf{T}\left(\mathbf{P}_k^T\mathbf{P}_k\right)\mathbf{D}_k\mathbf{P}_k^T = \mathbf{T}\mathbf{P}_k^T\mathbf{R}_k^2 \qquad \text{for} \quad k = (1, \ldots, K) \\ \mathbf{X}_k &= \mathbf{T}\mathbf{D}_k\left(\mathbf{T}^T\mathbf{T}\right)\mathbf{P}_k^T = \mathbf{S}_k\mathbf{T}\mathbf{P}_k^T \qquad \text{for} \quad k = (1, \ldots, K)\ , \end{aligned} \tag{13}$$

with $\mathbf{R}_k^2 = \mathbf{P}_k\mathbf{D}_k\mathbf{P}_k^T$ the square correlation matrix between (standardized) block-variables and $\mathbf{S}_k = \mathbf{T}\mathbf{D}_k\mathbf{T}^T$ the block scalar product matrix of observations into common space. The correlations between block-variables ($\mathbf{R}_k$) — mainly interpreted by users — are carried by observation-structure ($\mathbf{S}_k$).

Simulation model. Few studies have been devoted to multiblock data simulation and canonical factorization highlights a model — common to several widely-used methods — that makes it possible. A simulation procedure of controlled observation-structure — associated with given components sought by canonical factorization — is proposed to compare multiblock methods. Each block is simulated separately, with a given observation-structure chosen identical or different from other blocks (e.g., separation into more or less separated clusters, noise). For each block, components $\mathbf{T}_k$ are simulated with a given observation-structure and a given eigenvalues $\mathbf{D}_k$. From $\mathbf{D}_k$, a correlation-matrix $\mathbf{R}_k$ is derived [30]. From Eq. (13), $\mathbf{P}_k$ loadings are derived from eigenvectors of $\mathbf{R}_k^2 = \mathbf{P}_k\mathbf{D}_k\mathbf{P}_k^T$. Finally, data are computed with $\mathbf{X}_k = \mathbf{T}_k\mathbf{D}_k\mathbf{P}_k^T$. The K blocks are simulated in the same way — each with a given controlled structure — then $\mathbf{X} = [\mathbf{X}_1|\ldots|\mathbf{X}_K]$ is obtained.

Comparison criteria. Multiblock methods were compared with three criteria. (i) Since common components $\mathbf{T}$ were comparable but might differ from one method (M_1) to another (M_2), they were compared pairwise by the absolute value of the correlation coefficient $\rho^{(h)} = |\,\mathrm{cor}(\mathbf{t}_{M_1}^{(h)}, \mathbf{t}_{M_2}^{(h)})|$ for a given dimension h. The closer the $\rho^{(h)}$ value was to 1 (or 0), the more similar (or different) the components were. (ii) Although common components may differ across methods, block-variance explained by them was a appropriate criterion to measure how components partition variance of blocks along dimensions. The second criterion came from diagonal elements of $\mathbf{D}_k^{(h)2} = \|\mathbf{X}_k^T\mathbf{t}^{(h)}\|^2$. This highlights the direction of block observation-structure on common structure $\mathbf{T}$. (iii) The third criterion measures whether simulated observation-structure was recalled by components $\mathbf{T}$. For this purpose, K adjusted-Rand index [31] was computed, for each block k and dimension h ($\mathrm{ARI}_k^{(h)}$), between expected (= simulated) and observed observation-structures. The observed structure comes from a K-means clustering applied to each component $\mathbf{t}^{(h)}$. Among the K computed ($\mathrm{ARI}_1^{(h)}, \ldots, \mathrm{ARI}_K^{(h)}$), the maximum value was kept ($\mathrm{ARI}^{(h)}$). The closer the ARI value was to 1 (or 0), the more the method recovered (or did not recover) expected observation-structure.

3. Application

The analyses were performed using R [32]. CPCA, MFA and STATIS come from the SVD of appropriately scaled (eventually block-scaled) matrix. MCOA came from the `ade4` package with 'mcoa' function [33] and CCSWA from the `RGCCA` package with `rgcca` function and 'hpca' options [29]. Simulated data were obtained from `monte` function (`fungible` package; [34]. The ARI criterion was computed with `adjustedRandIndex` function of `mclust` package [35] associated with `kmeans` function (`stats` package). The choice of optimal block-clustering was obtained from the most frequent results among 30 indices that evaluate clustering performance between 2 and 10 clusters with `NbClust` package [36]. The PCA plots were obtained with the `FactoMineR` [37] and `factoextra` packages [38].

3.1. Simulation study

The goal was to explain similarities and differences between multiblock methods according to three key questions for users. How do methods behave in case of noise-blocks (scenario **S1**)? In case of blocks with common and specific observation-structures (scenario **S2**)? In case of blocks with each specific observation-structures (scenario **S3**)? CPCA, MCOA and CCSWA were used without any block-scaling, whereas MFA and STATIS were specifically scaled. To measure ability of multiblock methods to explore data structure, components were simulated from random data and labelled 'RANDOM'. The simulated data consisted of $N = 90$ observations measured on $J = 35$ (standardized) variables organized in $K = 4$ blocks with $J_k = (10, 10, 10, 5)$. The **S1** data were composed of a block with a specific observation-structure ($\mathbf{X}_1$ with three well-separated clusters of 30 observations each, 80% of this structure being spanned on the first dimension and 20% on the second one) and three blocks of noise. The **S2** data consisted of two blocks with the same observation-structure ($\mathbf{X}_1$ and $\mathbf{X}_2$) and two blocks with each a different structure. The **S3** data consisted of four blocks each with a different observation-structure. For the latter two scenarios, the four structures consisted of three well-separated clusters each of 30 observations, with 80% (or 75%, 70%, 65%) of this structure being spanned on the first dimension and 20% (or 25%, 30%, 35%) on the second one. These values were chosen to fix the order of blocks on components. The result of an additional scenario (how do methods behave when all blocks share the same observation-structure) is shown in Sup. Mat. A.2. For each scenario, 50 data were simulated and mean values of criteria were given.

3.1.1. S1: How do multiblock methods behave in case of noise-blocks?

The block-PCAs in Fig. 1(A) illustrate the structure of $\mathbf{X}_1$ in three clusters and noise structures of other blocks. Fig. 1(B) shows that multiblock methods, especially CPCA, MCOA, STATIS and CCSWA, perform better than the RANDOM method. The difference between CPCA and MCOA — due to deflation procedure — is shown in Fig. 1(C) (mean$\rho = 1$ for the first dim., 0.24 for the fourth dim.). The $\mathbf{X}_1$ variance is recovered on the first dimension, the other block-variances are shifted to higher-order dimensions (Fig. 1(B)). CPCA, MCOA, STATIS and CCSWA find the same first component (mean$\rho = 0.99$–1; Fig. 1(C)) and recover informative structure of $\mathbf{X}_1$ on this dimension (meanARI = 0.83–0.88; Fig. 1(D)). MFA does not recover the $\mathbf{X}_1$ structure on this dimension (meanARI = 0.12). This method is disadvantaged for this noise-scenario by its block-scaling which gives less importance to block(s) containing information on the first dimension (here, $\mathbf{X}_1$ with $\lambda_1^{(1)} = 79.4\%$; Fig. 1(A)).

3.1.2. S2: How do multiblock methods behave in case of blocks with common and specific observation-structures?

The block-PCAs in Fig. 2(A) illustrate the common structure of $\mathbf{X}_1$ and $\mathbf{X}_2$ in three clusters ($ARI(\mathbf{X}_1, \mathbf{X}_2) = 1$) and the specific structures of the two other blocks (e.g., ($ARI(\mathbf{X}_1, \mathbf{X}_3) \approx 0.01$). Fig. 2(B) shows that all methods recover common structure of $\mathbf{X}_1$ and $\mathbf{X}_2$ (MeanARI = 0.93–0.95; Fig. 2(D)) with the same first component (Mean$\rho = 0.99$–1; Fig. 2(C)). However, there are differences for higher-order dimensions, especially for STATIS and CCSWA. According to its block-scaling stable along dimensions, STATIS gives importance to $\mathbf{X}_1$ and $\mathbf{X}_2$ on all dimensions (Fig. 2(B)), the method being only interested in what is common to blocks. Conversely and according to its block-scaling which can vary across dimensions, CCSWA highlights what is common ($\mathbf{X}_1$ and $\mathbf{X}_2$ on the first dimension, then focuses on other specific block-structures $\mathbf{X}_3$ then $\mathbf{X}_4$) on higher-order dimensions. Consequently, CCSWA recovers informative structure of blocks along dimensions (meanARI = 0.98–0.85-0.71 for the first three dimensions; Fig. 2(D)). The same conclusions were obtained when clusters are slightly separated or when block-cluster sizes differ.

3.1.3. S3: How do multiblock methods behave in case of blocks with each specific observation-structures?

The block-PCAs in Fig. 3(A) illustrate the specific structures of all blocks. The multiblock methods follow different strategies here. Due to its specific block-scaling, MFA includes the $\mathbf{X}_4$ block from the first dimension. Because of its rank specificity, MCOA concentrates information on the first dimension. The other methods partition the four block-variance along all dimensions with different components, hence different strategies.

3.2. Exploratory data integration for food data

3.2.1. Data, pre-processing and aim

The data came from the 'AlimaSSenS' project [39], which aimed to provide a range of foods adapted to chewing difficulties of elderly, combining pleasure and comfort of eating with nutritional efficiency. Data were collected on 73 subjects, aged between 67 and 87 years old, with poor or good dental health. The subjects evaluated three meat products (minced chicken, shredded beef and shredded chicken) twice. During these evaluations, 46 variables were collected and organized in three blocks. The $\mathbf{X}_1$ block concerned mouth comfort when eating products (28 variables). The $\mathbf{X}_2$ block came from *in vivo* aroma release and perception of foods from nasal space with PTR-ToF-MS (= Proton Transfer Reaction with Mass Spectrometry; 5 variables). The $\mathbf{X}_3$ block concerned oral food processing which includes mastication, salivation, bowl formation, enzyme digestion and swallowing of food bowls (13 variables). The concatenated data set $\mathbf{X}$ had 438 rows and 46 columns. Since variables were expressed in different units, they were standardized. For CPCA, MCOA and CCSWA, to avoid situations where blocks had a strong influence due to their size (= inertia), block-data were divided by square root of their number of variables. The aim was to explore relationships between mouth comfort ($\mathbf{X}_1$), aroma release ($\mathbf{X}_2$) and oral food processing ($\mathbf{X}_3$). For this purpose, exploratory multiblock methods were used, the challenge being to select the most suitable one.

3.2.2. Block description

The preliminary step of any exploratory multiblock method was to capture block-structure across variables (variance distribution along dimensions) and observations (clustering structure). PCA and K-means were applied to each block. The choice of optimal block-clustering was obtained from the most frequent results among 30 indices that evaluated clustering performance between 2 and 10 clusters. Results were presented in Sup. Mat. A.3. The 28 $\mathbf{X}_1$-variables (mouth comfort) were mainly clustered in the first dimension (23.6% of variance). The $\mathbf{X}_1$-observations were structured in 2 clusters of size (169, 269), separated from metadata-variables *Dentition* and *Age* and block-variables *times*, *swallow* and *incisive*. The 5 $\mathbf{X}_2$-variables (aroma release) were summarized on the first two dimensions (47.1% and 35.1% of variance). The $\mathbf{X}_2$-observations were structured in 3 clusters of size (124, 157, 157), separated from metadata-variable *Product* and block-variable *imax.Hex*. The 13 $\mathbf{X}_3$-variables (oral food processing) were summarized on the

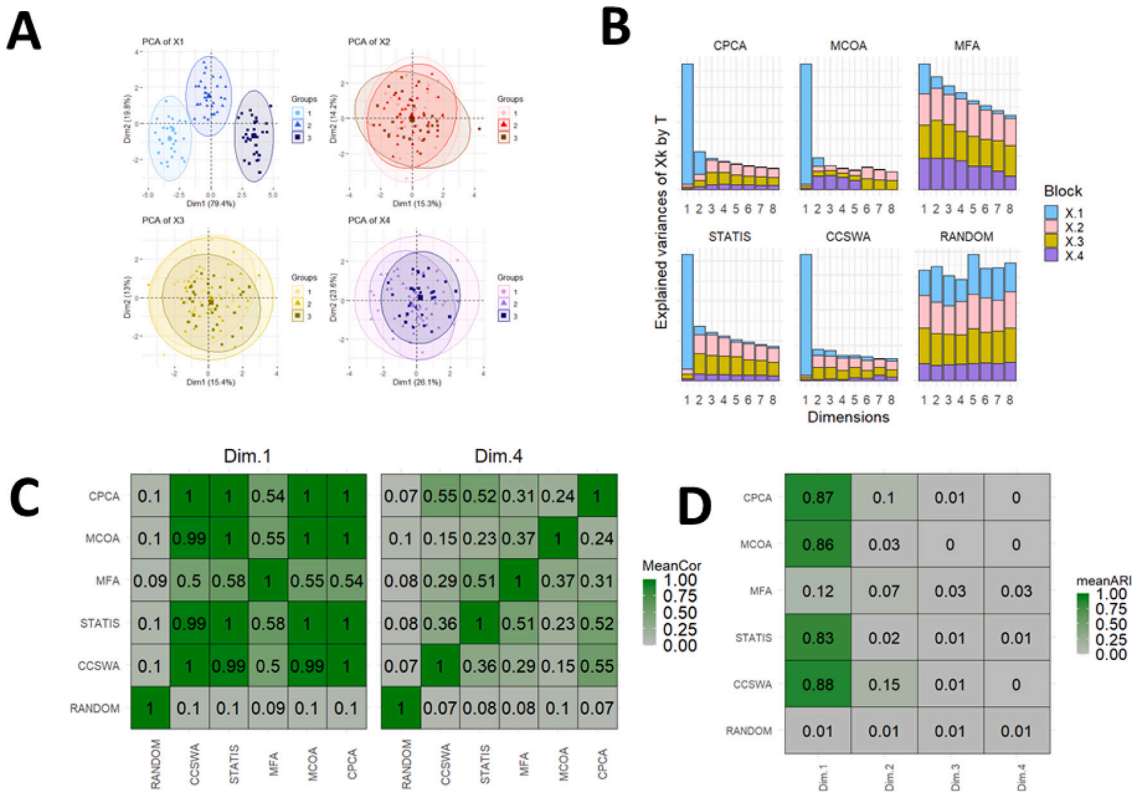


Fig. 1. How do multiblock methods behave in case of noise-blocks (**S1**)? **A**: Simulated blocks (block-PCA), **B**: Block-variances provided by components ($\mathbf{D}_k^2$), **C**: Common-components' similarity ($\rho^{(h)}$), **D**: Ability to recover the structure (ARI).

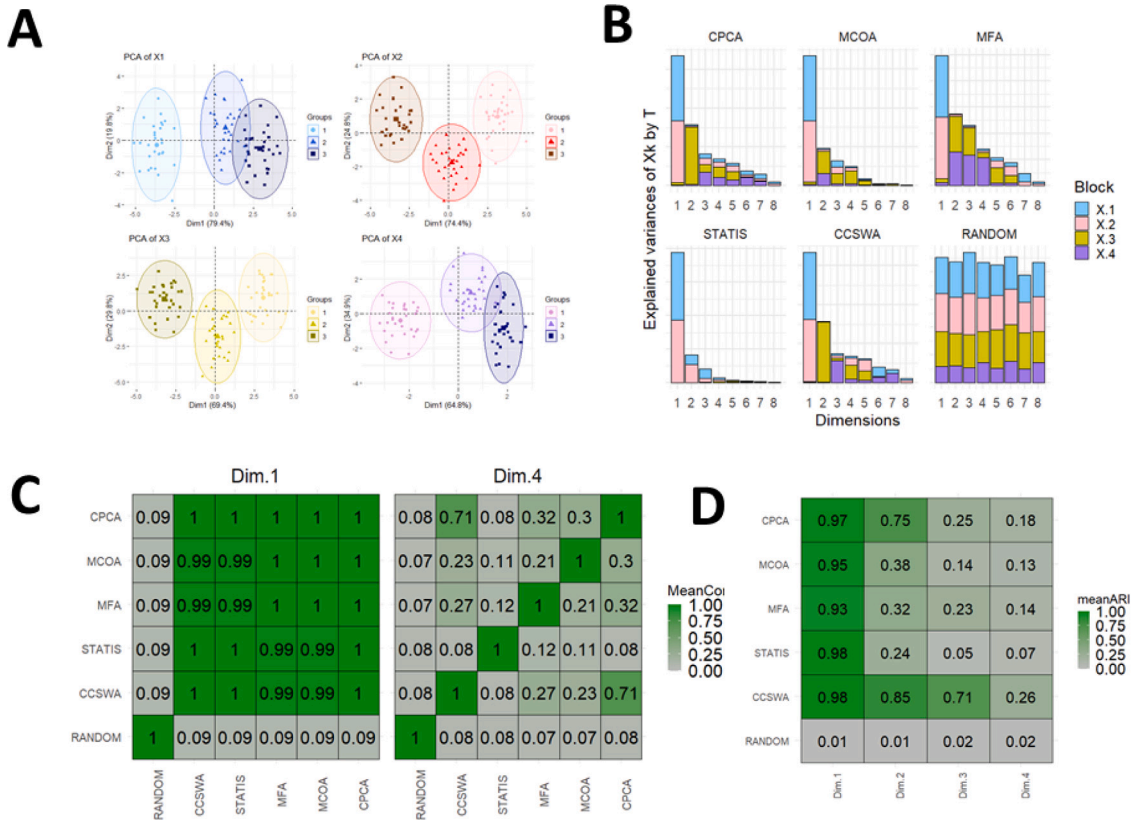


Fig. 2. How do multiblock methods behave in case of blocks with both common and specific observation-structures (**S2**)? **A**: Simulated blocks (block-PCA), **B**: Block-variances provided by components ($\mathbf{D}_k^2$), **C**: Common-components' similarity ($\rho^{(h)}$), **D**: Ability to recover the structure (ARI).

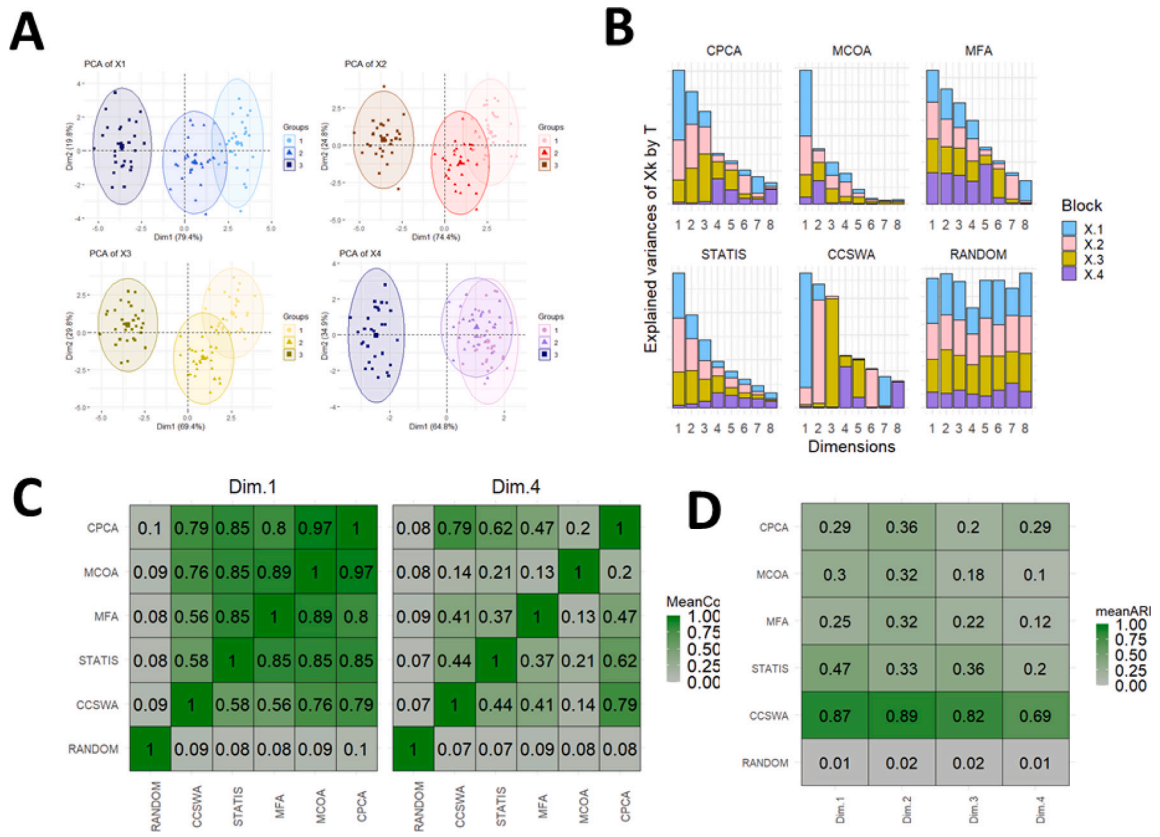


Fig. 3. How do multiblock methods behave in case of blocks with all specific observation-structures (**S3**)? **A**: Simulated blocks (block-PCA), **B**: Block-variances provided by components ($\mathbf{D}_k^2$), **C**: Common-components' similarity ($\rho^{(h)}$), **D**: Ability to recover the structure (ARI).

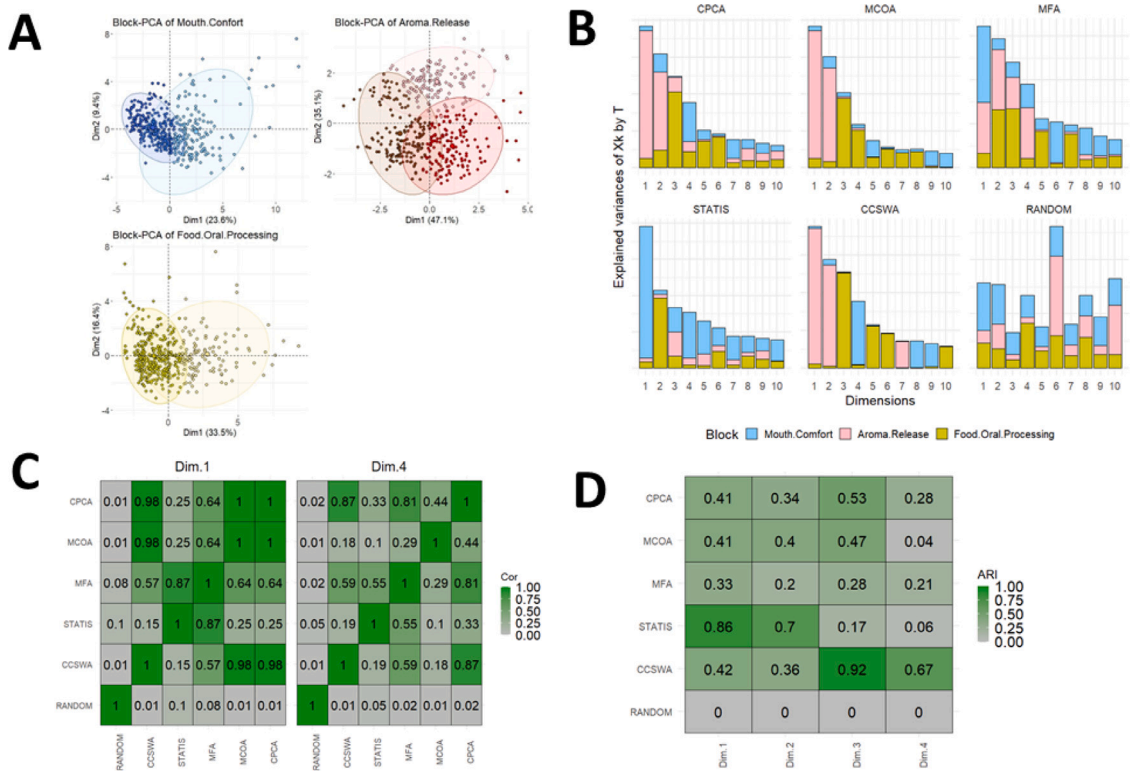


Fig. 4. Meat data - Comparison of multiblock methods. **A**: Block-PCA, **B**: Block-variances provided by components ($\mathbf{D}_k^2$), **C**: Common-components' similarity ($\rho^{(h)}$), **D**: Ability to recover the structure (ARI).

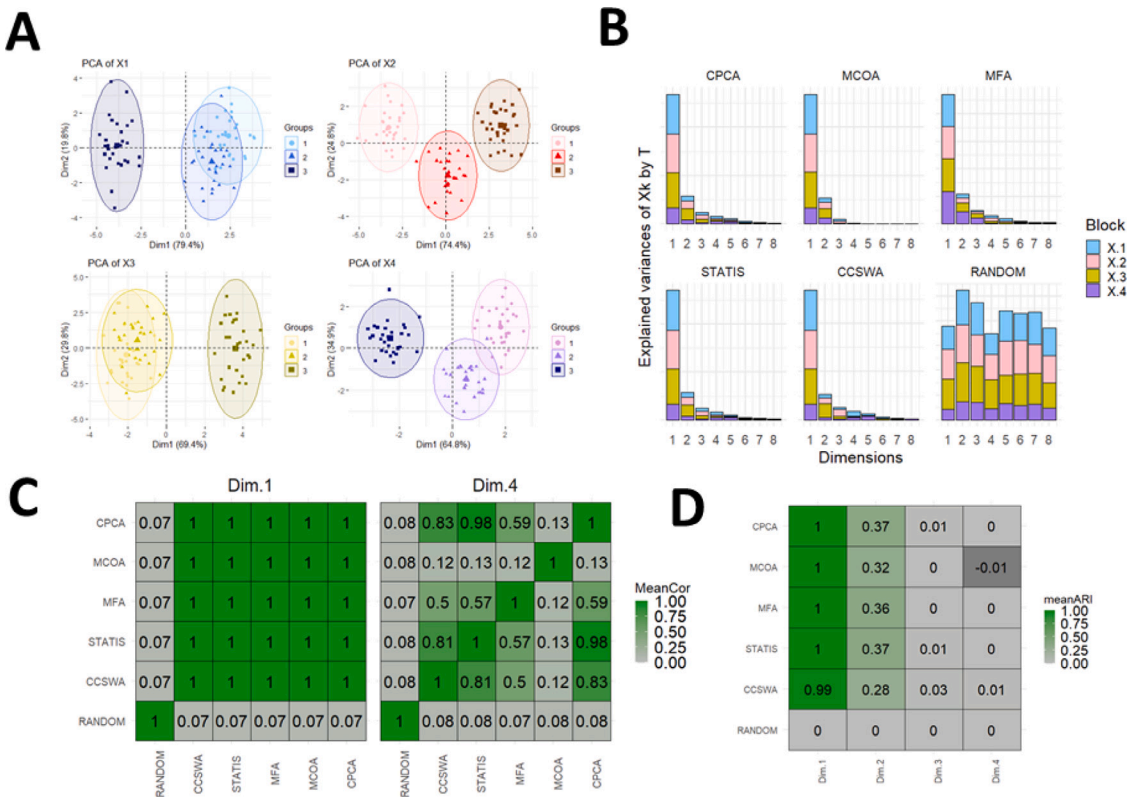


Fig. 5. How do methods behave when all blocks have a common structure (**S1**)? **A**: Simulated blocks (block-PCA), **B**: Block-variances provided by components ($\mathbf{D}_k^2$), **C**: Common-components' similarity ($\rho^{(h)}$), **D**: Ability to recover the structure (ARI).

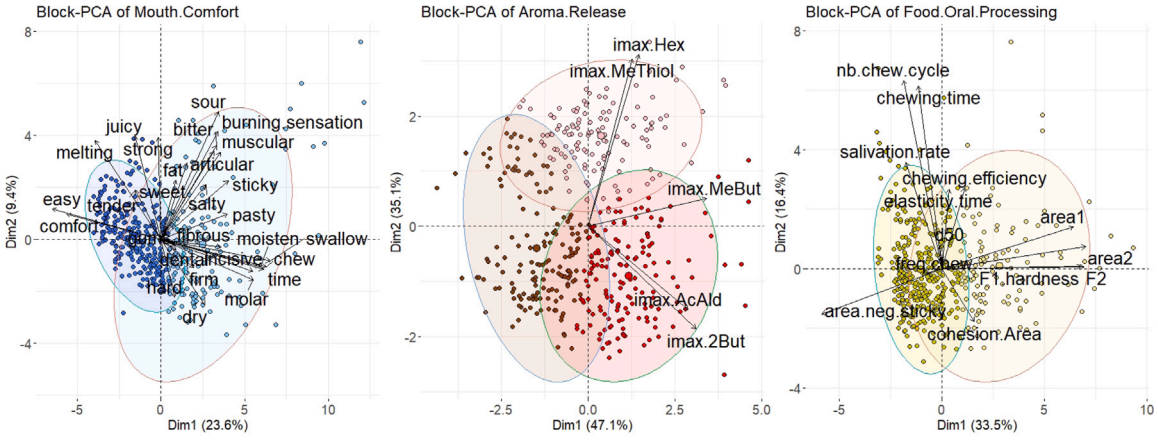


Fig. 6. Meat data - Block-PCA biplots coloured according to their observation-structure.

first component (33.5% of variance). The $\mathbf{X}_3$-observations were structured in 2 clusters of size (117, 321), separated from metadata-variables *Age* and *Dentition* and block-variables *area2*, *F2*, *area1*, *F1.hardness*, *area.neg.sticky*. These block-clusters were not related with each others: ARI(Clust.X_1, Clust.X_2) = 0.0145 and RV($\mathbf{X}_1$, $\mathbf{X}_2$) = 0.06; ARI(Clust.X_1, Clust.X_3) = 0.000 and RV($\mathbf{X}_1$, $\mathbf{X}_3$) = 0.05; ARI(Clust.X_2, Clust.X_3) = 0.006 and RV($\mathbf{X}_2$, $\mathbf{X}_3$) = 0.04. Thus, each block had a specific structure (**S3** scenario).

3.2.3. Multiblock method comparison and strategies

From Fig. 4(B) and according to its block-scaling, MFA gave more importance to mouth comfort block ($\mathbf{X}_1$) than CPCA or MCOA. STATIS tried to highlight a common structure that do not clearly exist. For CPCA, MCOA and CCSWA, the main information came from aroma release ($\mathbf{X}_2$) (Dim. 1 and 2) then food oral processing ($\mathbf{X}_3$) (Dim. 3). Indeed, $\mathbf{X}_2$ observation-structure hold on two dimensions (82.2% of its block-variance; block-PCA) and $\mathbf{X}_3$ on the third one (33.5% of its block-variance; block-PCA). CCSWA made different choices and highlighted aroma release ($\mathbf{X}_2$) on the first dimension, mouth comfort ($\mathbf{X}_1$) on the second dimension and oral food processing ($\mathbf{X}_3$) on the third one. Since our goal was to study observation structure of blocks — with equal importance in analysis — but also relationships between variables, CPCA was chosen.

3.2.4. CPCA interpretation

The CPCA results were shown in Sup. Mat. A.4 and A.5. The observation-structure of aroma release ($\mathbf{X}_2$) was supported by dimensions 1 and 2 (30.3% of variance), then that of oral food processing

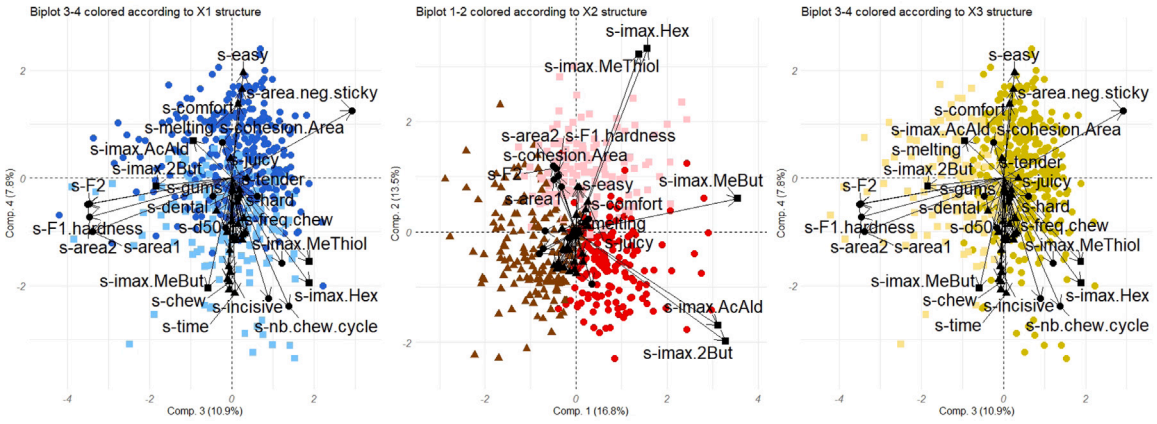


Fig. 7. Meat data - CPCA biplots for dimensions (1,2) and (3,4) coloured according to block observation-structure.

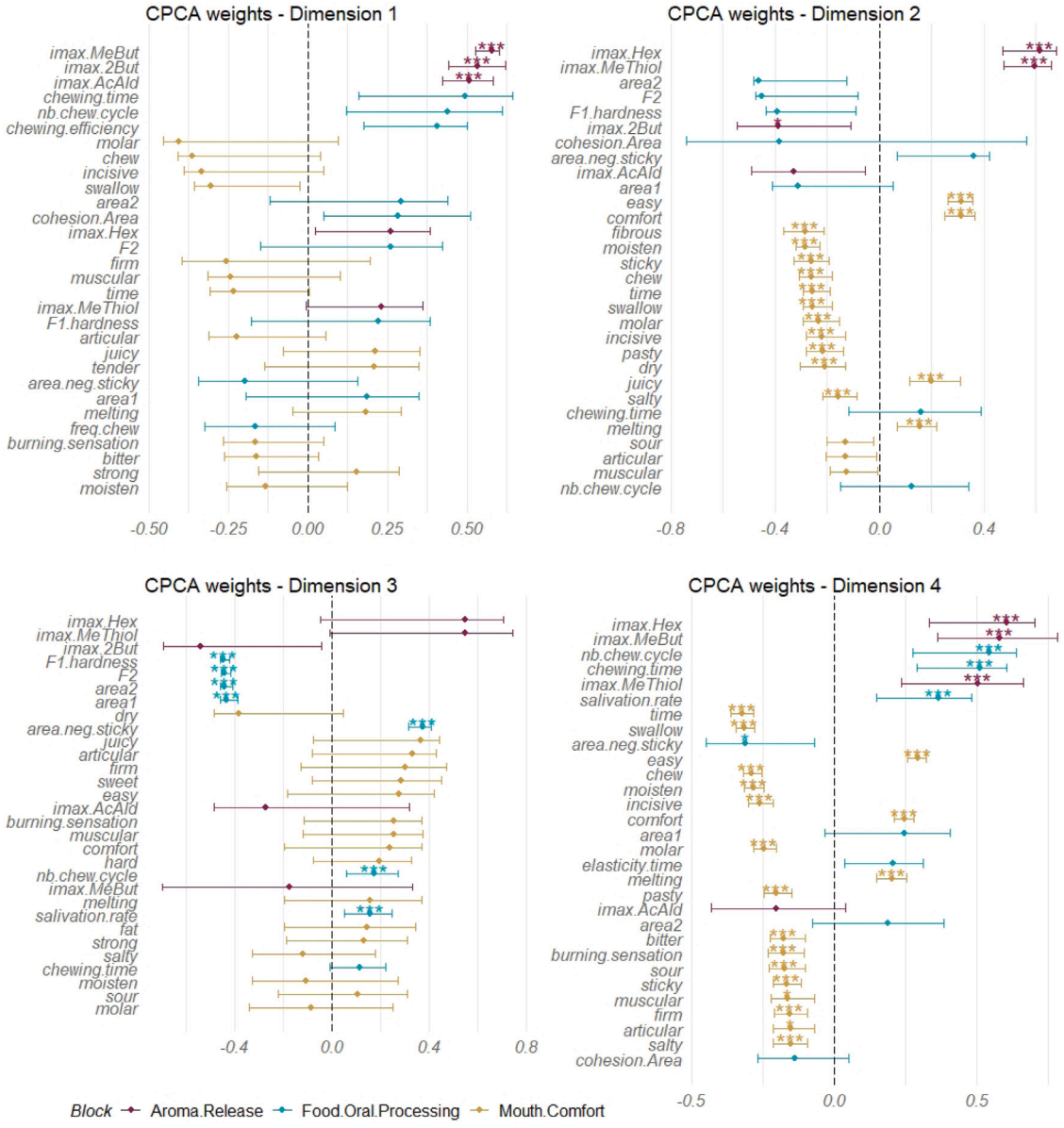


Fig. 8. Meat data - CPCA weights with 95% confidence intervals and significance (500 bootstrap simulations) for dimensions 1 to 4.

($\mathbf{X}_3$) by dimension 3 (10.9.% of variance) and the one of mouth comfort ($\mathbf{X}_1$) by dimension 4 (7.8% of variance). (i) The 1–2 CPCA components highlighted aroma release ($\mathbf{X}_2$) structure related to products ($\mathbf{X}_2$-Cluster 1 with *minced chicken*; $\mathbf{X}_2$-Cluster 2 with *shredded beef*; $\mathbf{X}_2$-Cluster 3 with *shredded chicken*). These $\mathbf{X}_2$-clusters were explained by five aroma release ($\mathbf{X}_2$) variables *imax.Hex*, *imax.2But*, *imax.AcAld*, *imax.MeThiol*, *imax.MeBut*, and to a lesser extent by five oral food processing ($\mathbf{X}_3$) variables *chewing.time*, *nb.chew.cycle*, *chewing.efficiency*, *Area.2* and *F2*, and 15 mouth comfort ($\mathbf{X}_1$) ones, *easy* and *comfort* among others. (ii) The third CPCA component highlighted oral food processing ($\mathbf{X}_3$) structure related to age and dentition ($\mathbf{X}_3$-Cluster 1 with > 80 years old and poor dental health; $\mathbf{X}_3$-Cluster 2 with <70 years old and good dental health). This $\mathbf{X}_3$-structure was mainly explained by aroma release ($\mathbf{X}_2$) variables (*imax.Hex*, *imax.MeThiol* and *imax.2But*) and oral food processing ($\mathbf{X}_3$) variables such as *F1.hardness*, *F2*, *Area1* and *Area2*.(iii) The fourth CPCA component highlighted mouth comfort ($\mathbf{X}_1$) structure related to age and dentition ($\mathbf{X}_1$-Cluster 1 with 70–80 years old and poor dental health; $\mathbf{X}_1$-Cluster 2 with <70 years old and good dental health). This $\mathbf{X}_1$-structure was explained by variables related to all blocks: all the mouth comfort variables ($\mathbf{X}_1$) in particular *times* and *swallow*, three aroma release ($\mathbf{X}_2$) variables in particular *imax.Hex* and *imax.MeBut*, and four oral food processing ($\mathbf{X}_3$) variables, in particular *nb.chew.cycle* and *chewing.time*.

4. Conclusion and perspectives

Our aim was to provide a comprehensive and unified framework — based on homogeneous outputs similar to those of PCA — to compare and explain strategies of multiblock exploratory methods. Many methods have been proposed independently and users have been lost in choosing the appropriate one. The data decomposition by canonical factorization was introduced and applied to widely-used methods CPCA, MCOA, MFA, STATIS and CCSWA (also known as COMDIM, ACCPS, HPCA). This factorization extracts parameters that highlight strategy adopted by methods. The methods have been compared on simulated (resp. real) data whose structure is controlled (resp. known). Theoretical and practical results show that block-structure must be explored beforehand, e.g., with K block-PCA, in order to answer two questions before choosing method. The first question concerns the number of block-variables and block-variance distribution along dimensions, which affects block-scaling (no block-scaling/block-scaling with inertia/MFA-like block-scaling). The second issue affects observation-structure within and between blocks, which impacts the choice of the method. In short, CPCA or MCOA mix common and specific information, STATIS emphasizes only common structure only while CCSWA focuses on specific information. Methods and comparison tools were available on R (available code in Supplementary Material), `Matlab/Octave` ('MultiBlock toolbox for Chemometrics' `MB4Chem` package; https://forgemia.inra.fr/chemhouse/octave/mb4chem) or with a `Galaxy` web application based on the `MB4Chem` package (https://vm-chemflow-francegrille.eu).

Although solution proposed by canonical factorization was original and appropriate, further theoretical and empirical work need to be done. For instance, the proposed matrix decomposition could be improved by constraints related to B-factorization of Simultaneous Component Analysis [14,40,41]. The effect of block-scaling was considered here as a block-weighted PCA, but could be theoretically investigated with a generalized eigenvalue problem [42]. Canonical factorization could also be applied to multiblock supervised methods, insofar as their criterion can be written as a PCA-like one (e.g., multiblock PLS, multiblock PCAIV as special cases of rGCCA). From a practical point of view, more complex block-clusters could be simulated (e.g., different direction and/or shapes across clusters, clusters visible only on higher order dimensions). The common observation-structure between blocks could be investigated with consensus clustering [43]. The proposed approach allows us to integrate other multiblock methods, such as JIVE [44] devoted to exploration of common and specific block-structures. This researches together with the increase in data volume and complexity will help to make use of exploratory multiblock methods more popular.

Table 1
Exploratory multiblock methods as special cases of rGCCA with a super-block [29].

Multiblock method	Criterion & constraints	Block-scaling	Package & option
CPCA [5]	$\sum_k \text{cov}^2(\mathbf{X}_k\mathbf{w}_k, \mathbf{X}_{k+1}\mathbf{w}_{k+1})$ with $\|\mathbf{t}\| = \|\mathbf{w}_k\| = 1$	$1/\sqrt{J_k}$	RGCCA, 'cpca-2'
MCOA [9]	$\sum_k \text{cov}^2(\mathbf{X}_k\mathbf{w}_k, \mathbf{X}_{k+1}\mathbf{w}_{k+1})$ with $\|\mathbf{t}\| = \|\mathbf{w}_k\| = 1$	$1/\sqrt{J_k}$	RGCCA, 'mcoa'
MFA [7]	$\sum_k \text{cov}^2(\mathbf{X}_k\mathbf{w}_k, \mathbf{X}_{k+1}\mathbf{w}_{k+1})$ with $\|\mathbf{t}\| = \|\mathbf{w}_k\| = 1$	$1/\sqrt{\lambda_k^{(1)}}$	RGCCA, 'mfa'
STATIS [8]	$\sum_k \text{cov}^2(\mathbf{X}_k\mathbf{w}_k, \mathbf{X}_{k+1}\mathbf{w}_{k+1})$ with $\|\mathbf{t}\| = \|\mathbf{w}_k\| = 1$	$\sqrt{\alpha_k}$	ade4, 'statis'
CCSWA [12]	$\sum_k \text{cov}^4(\mathbf{X}_k\mathbf{w}_k, \mathbf{X}_{k+1}\mathbf{w}_{k+1})$ with $\|\mathbf{t}\| = \|\mathbf{w}_k\| = 1$	$1/\sqrt{J_k}$	RGCCA, 'hpca'

CRediT authorship contribution statement

Stéphanie Bougeard: Writing – original draft, Software, Methodology. **Caroline Peltier:** Validation, Software, Methodology. **Benoit Jaillais:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Jean-Claude Boulet:** Writing – review & editing, Validation, Software, Investigation. **Mohamed Hanafi:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A

A.1. Multiblock methods as special cases of rGCCA

See Table 1.

A.2. How do methods behave when all blocks share the same structure?

All multiblock methods recover observation-structure on the first two dimensions and in the same way, which is not the case for the RANDOM method (see Figs. 5–8 and Table 1).

A.3. Meat data - block-PCA biplots coloured according to their observation-structure

See Fig. 6.

A.4. Meat data - CPCA biplots coloured according to block observation-structure

See Fig. 7.

A.5. Meat data - CPCA weights of meat data

See Fig. 8.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.chemolab.2024.105240.

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